

## PERRYTON OCHILTREE COUNTY AP, TX, USA

WMO: 720315

Lat: 36.414N

Lon: 100.750W

Elev: 2918

StdP: 13.21

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03045

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 9.5        | 14.3 | 0.1                           | 6.1 | 16.8 | 4.1 | 7.6 | 23.4 | 33.8                  | 47.8 | 31.1 | 47.4 | 10.5                  | 0    | 0.636 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 26.3                   | 101.6           | 67.3 | 99.2 | 67.4 | 96.7 | 67.5 | 73.0                | 88.0 | 71.8 | 87.2 | 70.8 | 86.2 | 16.0                 | 190  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 69.3                            | 120.3 | 77.5 | 68.1 | 115.1 | 76.7 | 66.7 | 109.5 | 75.9 | 38.6          | 87.9 | 37.5 | 87.7 | 36.6 | 85.8 | 77.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 31.6          | 27.2 | 24.6 | 1.9                        | 106.1 | 5.1                | 3.3 | -1.8                                               | 108.5 | -4.8       | 110.4 | -7.7       | 112.3 | -11.4      | 114.7 |    |    |
| (5)               |      |      | 0.8                        | 74.7  | 4.7                | 1.6 | -2.6                                               | 75.9  | -5.4       | 76.8  | -8.0       | 77.7  | -11.5      | 78.8  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 58.4        | 36.8  | 38.8  | 49.0  | 56.6  | 66.1  | 77.3  | 80.7  | 78.9  | 72.2  | 59.1  | 47.3  | 36.8  |  |  |
|                                                                        | DBStd       | 18.19       | 9.82  | 11.36 | 9.97  | 9.35  | 8.80  | 6.18  | 5.54  | 5.50  | 7.17  | 9.33  | 9.60  | 9.78  |  |  |
|                                                                        | HDD50       | 1543        | 420   | 339   | 138   | 40    | 4     | 0     | 0     | 0     | 0     | 26    | 160   | 416   |  |  |
|                                                                        | HDD65       | 4142        | 875   | 734   | 502   | 280   | 95    | 2     | 0     | 0     | 21    | 227   | 533   | 873   |  |  |
|                                                                        | CDD50       | 4607        | 9     | 25    | 106   | 237   | 503   | 820   | 953   | 895   | 665   | 307   | 79    | 8     |  |  |
|                                                                        | CDD65       | 1736        | 0     | 1     | 6     | 28    | 129   | 372   | 488   | 431   | 236   | 43    | 2     | 0     |  |  |
|                                                                        | CDH74       | 21554       | 4     | 43    | 174   | 582   | 1763  | 4598  | 6186  | 4823  | 2645  | 647   | 87    | 2     |  |  |
|                                                                        | CDH80       | 11451       | 0     | 10    | 46    | 218   | 809   | 2521  | 3642  | 2672  | 1280  | 237   | 16    | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 11.8        | 11.3  | 11.9  | 12.6  | 13.8  | 12.0  | 12.8  | 11.0  | 10.0  | 11.3  | 11.6  | 11.6  | 11.2  |  |  |
| (15) Precipitation                                                     | PrecAvg     | 21.7        | 0.6   | 0.6   | 1.5   | 1.8   | 2.8   | 3.2   | 3.1   | 2.9   | 1.7   | 1.8   | 0.9   | 0.9   |  |  |
|                                                                        | PrecMax     | 26.8        | 2.0   | 1.6   | 3.9   | 6.1   | 5.3   | 6.0   | 7.8   | 5.7   | 6.1   | 5.7   | 4.1   | 3.7   |  |  |
|                                                                        | PrecMin     | 15.6        | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   | 0.2   | 0.9   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   |  |  |
|                                                                        | PrecStd     | 3.0         | 0.6   | 0.5   | 1.1   | 1.7   | 1.1   | 1.7   | 1.9   | 1.4   | 1.4   | 1.6   | 1.0   | 0.9   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 72.4  | 80.1  | 85.4  | 92.1  | 98.6  | 105.9 | 105.3 | 103.9 | 99.4  | 92.5  | 82.0  | 69.7  |  |  |
|                                                                        |             | MCWB        | 49.7  | 51.4  | 55.1  | 57.4  | 61.2  | 64.8  | 67.6  | 68.1  | 65.6  | 62.2  | 54.8  | 49.7  |  |  |
|                                                                        | 2%          | DB          | 65.8  | 72.0  | 79.2  | 86.3  | 92.9  | 100.0 | 101.6 | 99.9  | 95.5  | 86.3  | 75.4  | 63.4  |  |  |
|                                                                        |             | MCWB        | 46.3  | 50.0  | 54.1  | 57.4  | 62.0  | 66.5  | 68.2  | 68.1  | 65.7  | 62.1  | 53.8  | 46.9  |  |  |
|                                                                        | 5%          | DB          | 59.3  | 65.0  | 73.0  | 81.1  | 89.2  | 96.8  | 99.3  | 97.2  | 91.4  | 81.4  | 70.1  | 58.3  |  |  |
|                                                                        |             | MCWB        | 43.6  | 46.7  | 52.4  | 56.9  | 62.2  | 67.3  | 68.1  | 68.6  | 65.9  | 59.9  | 51.5  | 44.6  |  |  |
|                                                                        | 10%         | DB          | 53.9  | 57.7  | 67.9  | 75.3  | 84.2  | 93.0  | 96.9  | 93.4  | 88.0  | 76.1  | 63.9  | 53.3  |  |  |
|                                                                        |             | MCWB        | 41.4  | 43.8  | 51.2  | 56.0  | 62.2  | 67.5  | 68.3  | 68.6  | 65.5  | 58.9  | 49.2  | 42.5  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 51.1  | 54.9  | 62.4  | 64.8  | 71.1  | 74.0  | 74.7  | 73.9  | 72.6  | 69.0  | 59.3  | 53.3  |  |  |
|                                                                        |             | MCDB        | 69.3  | 70.9  | 73.5  | 76.2  | 82.9  | 89.7  | 89.5  | 88.4  | 85.9  | 79.8  | 68.3  | 60.6  |  |  |
|                                                                        | 2%          | WB          | 47.8  | 50.8  | 58.9  | 62.3  | 68.9  | 72.3  | 73.1  | 72.4  | 70.3  | 66.3  | 55.8  | 48.7  |  |  |
|                                                                        |             | MCDB        | 63.0  | 69.3  | 69.1  | 73.8  | 80.4  | 88.5  | 88.1  | 87.3  | 83.8  | 76.7  | 69.1  | 60.4  |  |  |
|                                                                        | 5%          | WB          | 44.7  | 47.8  | 55.8  | 59.9  | 66.9  | 71.0  | 71.7  | 71.3  | 69.0  | 63.9  | 53.2  | 45.5  |  |  |
|                                                                        |             | MCDB        | 58.3  | 64.5  | 67.2  | 72.4  | 78.8  | 87.1  | 87.8  | 86.3  | 82.6  | 74.2  | 66.5  | 56.7  |  |  |
|                                                                        | 10%         | WB          | 41.7  | 44.5  | 52.8  | 57.6  | 64.8  | 69.8  | 70.6  | 70.4  | 67.9  | 61.3  | 50.7  | 42.5  |  |  |
|                                                                        |             | MCDB        | 52.9  | 57.3  | 66.8  | 71.7  | 77.9  | 85.2  | 86.8  | 85.0  | 81.0  | 71.9  | 63.6  | 52.7  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 25.8  | 26.4  | 28.2  | 28.0  | 26.8  | 26.8  | 26.3  | 25.4  | 26.3  | 26.6  | 27.1  | 23.8  |  |  |
|                                                                        |             | MCDBR       | 35.3  | 39.0  | 35.9  | 37.5  | 34.2  | 31.8  | 30.6  | 30.4  | 31.2  | 33.9  | 34.5  | 32.2  |  |  |
|                                                                        | 5% WB       | MCWBR       | 19.9  | 20.1  | 17.4  | 16.0  | 12.8  | 8.9   | 7.6   | 7.5   | 9.6   | 14.3  | 16.8  | 18.5  |  |  |
|                                                                        |             | MCWBR       | 33.3  | 37.0  | 29.6  | 27.3  | 26.3  | 26.5  | 25.9  | 24.5  | 24.7  | 25.5  | 30.4  | 29.7  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.257 | 0.265 | 0.297 | 0.325 | 0.343 | 0.373 | 0.381 | 0.366 | 0.337 | 0.301 | 0.273 | 0.257 |  |  |
|                                                                        |             | taud        | 2.586 | 2.578 | 2.487 | 2.426 | 2.443 | 2.402 | 2.411 | 2.417 | 2.479 | 2.560 | 2.571 | 2.591 |  |  |
|                                                                        | Ebn at Noon |             | 305   | 314   | 310   | 304   | 298   | 287   | 284   | 287   | 292   | 295   | 295   | 298   |  |  |
|                                                                        |             | Edh at Noon | 23    | 26    | 32    | 36    | 36    | 37    | 37    | 36    | 32    | 27    | 23    | 22    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 916   | 1169  | 1557  | 1933  | 2185  | 2352  | 2295  | 2036  | 1733  | 1291  | 990   | 802   |  |  |
|                                                                        | RadStd      |             | 81    | 123   | 123   | 117   | 151   | 141   | 107   | 101   | 82    | 154   | 95    | 61    |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (47) | Regional (19 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page